Network Modeling and Analysis in Python

Today's goal • 1/3 items

Module 1
Connectivity in Networks

Module 2
Community Structure in Networks

Module 3
Network Generative Models

Module 4
Models of Diffusion in Networks

Basic Models of Diffusion in Networks (Part 1)

- Introduction to Diffusion in Networks
  Video • 7 min
- The SIS Model
  Video • 8 min
- The SIR Model
  Video • 6 min
- Knowledge check: Basic Models of Diffusion in Networks (Part 1)
  Practice Assignment • Grade: 100%

Basic Models of Diffusion in Networks (Part 2)

- The Independent Cascade Model
  Video • 5 min
- The Linear Threshold Model (Part 1)
  Video • 8 min
- The Linear Threshold Model (Part 2)
  Video • 5 min
- Knowledge Check: Basic Models of Diffusion in Networks (Part 2)
  Practice Assignment • Grade: 100%

The Influence Maximization Problem

- Honduras Social Networks
  Reading • 10 min
- Introduction to the Influence Maximization Problem
  Video • 3 min
- Properties of the Influence Maximization Problem
  Video • 14 min
- Greedy Algorithm for the Influence Maximization Problem
  Video • 4 min
- Heuristics and Nomination Strategy
  Video • 10 min
- Applications of Community Detection
  Video • 7 min
- Knowledge Check: The Influence Maximization Problem
  Practice Assignment • Grade: 100%

Module 4 Wrap-Up

- Demo: Diffusion NetworkX Tutorial

Optional Readings and Resources

0:10 / 1:50 2x

- D. Easley and J. Kleinberg, *Networks, crowds, and markets : reasoning about a highly connected world*. New York: Cambridge University Press, 2010. https://www.cs.cornell.edu/home/kleinber/networks-book/
  - Sections 19.1-19.3: Cascading Behavior in Networks
  - Chapter 21: Epidemics
- D. Kempe, J. Kleinberg, and É. Tardos, "Maximizing the spread of influence through a social network," *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining - KDD '03*, 2003, doi: https://doi.org/10.1145/956750.956769. (alternative link https://www.cs.cornell.edu/home/kleinber/kdd03-inf.pdf
- D. A. Kim *et al.*, "Social network targeting to maximise population behaviour change: a cluster randomised controlled trial," *The Lancet*, vol. 386, no. 9989, pp. 145–153, Jul. 2015, doi: https://doi.org/10.1016/s0140-6736(15)60095-2. (alternative link: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4638320/
- G. Nemhauser, L. Wolsey, M. Fisher. An analysis of the approximations for maximizing submodular set functions. Mathematical Programming, 14(1978), 265–29 https://doi.org/10.1007/BF01588971
- Book: Network Science by Albert-László Barabási
  - Chapter 10, Spreading phenomena https://networksciencebook.com/chapter/10#introduction10
- Newman, M. (2018). Networks. Oxford University PressChapter 16: Epidemics on Networks https://doi.org/10.1093/oso/9780198805090.003.0016

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